SHIZA KHAN

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EDUCATION & AWARDS

University of Waterloo **Waterloo, ON**
BASc. Honours Mechatronics Engineering *September 2026 – Present*

- **Expected Graduation:** April 2031.

Schulich Leader Foundation **National Level**
$120,000 Recipient of the Schulich Leader Scholarship *May 2026*

- Selected as a scholarship recipient for leadership, creativity, and entrepreneurial potential in STEM.

PROFESSIONAL EXPERIENCE

University of Waterloo's Rocketry Team **Waterloo, ON**
Firmware Department Member *September 2026 – Present*

- Read PCB schematics to identify GPIO connections and programmed an STM32-based board via ST-Link using the STM HAL library to blink an LED on and off.
- Debugged firmware upload failures to the circuit board by checking the ST-Link connection and finding a driver mismatch, resolving the issue by installing the correct driver.
- Developed an error-logging system (using definition files, ID strings, and a Python decoding script) designed to shrink log messages from 128 to 32 bytes, allowing 4x as many messages in the same bandwidth; now implementing it with other team members.

PROJECTS

Arduino Pong Game | Arduino UNO R4, C++ | GitHub *Sept 2026 – Oct 2026*

- Implemented a Pong game on an Arduino board using an OLED display and push buttons in a team of 2.
- Responsible for programming the game logic, including ball movement, collision detection, and a scoring system.
- Resolved inconsistent button input by testing buttons individually and writing a debouncing function.
- Debugged ball passing through paddles by logging ball and paddle coordinates over the serial monitor, identifying that high speeds let the ball skip past the paddle between frames, and fixing it by capping ball speed.

Smart Office Security System | Arduino UNO R4, C++ | GitHub *May 2026 – June 2026*

- Designed an Arduino-based circuit system simulating an office security and lighting system using sensor modules.
- Designed the circuit in Tinkercad before assembling and testing the physical prototype on a breadboard.
- Programmed a control loop to respond to user input and environmental conditions, enabling selectable lighting modes, intrusion detection, and alarm activation.
- Debugged false alarms from the sound sensor by logging analog readings over the serial monitor, determining its real output range in quiet and noisy conditions, and updating the control loop threshold with those sound levels.

Speaker Box Project | Seeeduino Nano, C++ *October 2026*

- Designed a speaker cabinet increasing the sound output of a 97.0 dB driver to 123.9 dB in a team of four.
- Conducted tests with defined objectives and controlled variables to determine the ideal parameters for maximum volume, then modified the provided control loop to include these parameters.
- Evaluated the team's design and development process in a post-mortem report, identifying strengths and areas of improvement for future projects.

TECHNICAL SKILLS

Electronics: Arduino UNO R4, Seeeduino Nano, STM32, Breadboard Prototyping, Control Logic, Circuit Debugging.

Programming Languages: C, C++, Python, SQL, Java, JavaScript.

Tools: AutoCAD, SolidWorks, Git/GitHub.

Soft Skills: Error Identification, Problem-Solving, Technical Documentation, Adaptability, Collaboration.